

ECON 701

Dynamic Information Design with Partial Commitment: How Much
Credibility Does Honest Disclosure from a Central Bank Actually
Need?

Hanshuo Ma

Summer 2026

- ➊ Introduction
- ➋ Motivation
- ➌ Dynamic extension
- ➍ Model Setup
- ➎ Full Commitment Benchmark
- ➏ Optimal policy
- ➐ Partial Commitment

Introduction

- The Persuasion Problem: the central bank wants to find an optimal information policy to maximize spending as much as possible throughout the business cycle.
- The Signals: interest-rate guidance takes the form of either a cut or a raise. A cut signals the normal state, and a raise signals recession.
- The household would like to spend in the normal state and save in recession, but it never observes the state.
- The Commitment Issue: the household knows that the central bank sticks to the announced policy only with probability $\lambda < 1$.
- Households spend on beliefs; the bank occasionally wants to break its own policy to steer FX.

Bayesian Persuasion Literature Review

- Canonical model: Kamenica & Gentzkow (2011)
- The receiver must make a decision, and that decision enters the sender's utility function directly.
- **State:** $\omega \in \Omega$, with common prior $\phi_0 \in \Delta(\Omega)$.
- **Player:** $i \in \{1, \dots, N\}$, where each player chooses an action $a_i \in A_i$.
- Each player's objective function is $v_i(a, \omega)$.
- **Designer:** chooses an experiment (μ, M) and commits to a signaling scheme $\mu : \Omega \rightarrow \Delta(M^N)$ before the state is realized.
- The designer's objective function is $v_0(a, \omega)$.

Main extensions

- Ely (2017): Dynamic Bayesian Persuasion with full commitment
- Min (2021): Static Bayesian Persuasion with partial commitment

	Belief path	Absorbing State
Ely (2017)	Monotonic	Yes
This paper	Ergodic process	No

Table: Main differences between the models

- Agent's belief path in Ely (2017): $\mu_t = 1 - e^{-\lambda t}$
- This paper's 2-state Markov chain:

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$$

Model Setup

- Two states $s_t \in S = \{N, R\}$
- Signals: the central bank discloses interest-rate guidance as a signal of the state — a cut signals the normal state, a raise signals recession.
- The household has two actions $a \in \{0, 1\}$.
- Stationary distribution: $\mu^* = (1 - \mu_R^*, \mu_R^*), a > b, \mu_R^* = \frac{a}{a+b} > \frac{1}{2}$
- The central bank's payoff depends on the household's interim belief:

$$u(\nu) = \begin{cases} 1 & \nu \leq p^* \\ 0 & \nu > p^* \end{cases}$$

- p^* is the threshold above which the household saves because it believes a recession is underway.
- The commitment is partial: with probability $1 - \lambda$, the central bank cheats and signals a rate cut regardless of the true state.

Full Commitment Benchmark Setup

- We are only interested in the case where $\mu_R^* > p^*$.

Theorem

(Ely 2017, Online Appendix 6.2, Lemma 1) Suppose that $I = (p^, 1]$ is absorbing under f , and that u is linear over I . Then V is also linear on I .*

- The belief map is derived directly from the Markov chain and is affine: $f(\mu) = a + (1 - a - b)\mu$.
- We need the Obfuscation Principle from Ely (2017) to fully characterize the dynamic optimization problem.

Theorem

Dynamic Obfuscation Principle: Any policy σ induces a stochastic process (μ_t, ν_t) satisfying

- 1 $\mathbb{E}(\nu_t \mid \mu_t) = \mu_t$
- 2 *There exists a linear map f such that $\mu_{t+1} = f(\nu_t)$*

Continue

- The central bank's expected payoff from such a policy σ is:

$$\mathbb{E} \sum_{t=0}^{\infty} \delta^t u(\nu_t)$$

where $u(\nu_t) := \sum_{s \in S} \nu_t(s) u(\nu_t, s)$

- Bellman equation:

$$V(\mu) = \max_{q: \mathbb{E}_q[\nu] = \mu} \mathbb{E}_q[(1 - \delta)u(\nu) + \delta V(f(\nu))]$$

- Notice how the value function is the concavification of a function that involves the value function itself.
- Thanks to Theorem 1 of Ely (2017), we know V exists as a unique fixed point of the Bellman operator.

Continue

Theorem

The optimal value function is characterized by the functional equation $V = \text{cav}[(1 - \delta)u + \delta(V \circ f)]$. In particular, V is the unique fixed point of the operator

$$TV = \text{cav}[(1 - \delta)u + \delta(V \circ f)],$$

which is a contraction mapping and therefore converges by iteration to V .

- Since V exists and is unique, and, by Lemma 1, is also affine, $V(\mu) = \bar{a}\mu + \bar{b}$ on $(p^*, 1]$. Thus we only need to find $V(p^*)$ and $V(1)$ to fully characterize V .

Continue

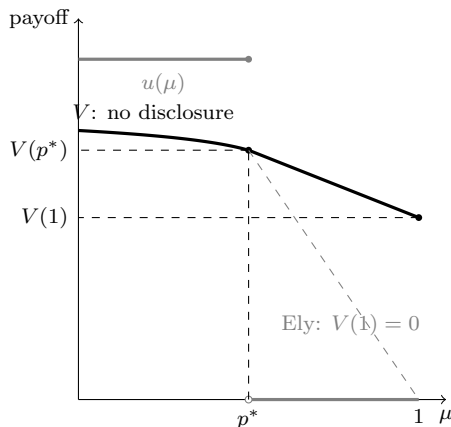


Figure: The affine rationing chord joining $(p^*, V(p^*))$ to $(1, V(1))$ — the kink at p^* is where disclosure begins. The vertical gap at $\mu = 1$ is the recurrence premium. Drawn to scale for $a = 0.3$, $b = 0.15$, $p^* = 0.5$, $\delta = 0.9$.

Continue

- Closed-form expressions of $V(1)$ and $V(p^*)$:

$$V(1) = \delta V(1 - b) = \delta \frac{b V(p^*) + (1 - b - p^*) V(1)}{1 - p^*} \quad (1)$$

$$\varphi := f(p^*) = a + (1 - a - b)p^* \in (p^*, 1) \quad (2)$$

$$V(p^*) = (1 - \delta) + \delta V(\varphi) \quad (3)$$

$$= (1 - \delta) + \delta \frac{(1 - \varphi) V(p^*) + (\varphi - p^*) V(1)}{1 - p^*} \quad (4)$$

- Two equations, two unknowns, solve for both:

$$V(1) = \frac{\delta b}{\Delta} V(p^*) , \quad \Delta = (1 - p^*)(1 - \delta) + \delta b \quad (5)$$

$$V(p^*) = \frac{(1 - \delta)(1 - p^*)}{(1 - p^*) - \delta(1 - \varphi) - \frac{\delta^2 b (\varphi - p^*)}{\Delta}} \quad (6)$$

$$\varphi - p^* = (a + b)(\mu_R^* - p^*) > 0 \quad (7)$$

Optimal policy under full commitment

- Phase A ($\mu \leq p^*$): While the belief is below the threshold, the bank issues uninformative cut guidance.
- Phase B ($f(p^*) = \varphi > p^*$): The bank issues an informative raise *only* in state R , at rate $\rho \in [0, 1]$, and never in state N .
- $\rho^*(\mu) = \frac{\mu - p^*}{\mu(1 - p^*)}$ is the disclosure rate: the probability that the bank issues a raise conditional on the state being R .
- Phase C ($\mu = 1$): Whenever a raise is issued, the message fully reveals state R and the posterior jumps to 1; the central bank then earns nothing until the belief drifts back down and re-enters Phase B.
- This delay-then-ration policy is also known as the greedy policy of Renault, Solan and Vieille (2017).

Partial Commitment

- Under partial commitment, the realised disclosure rate is driven down to $\lambda\rho$.
- Therefore, to remain credible, the central bank must still meet the required disclosure rate:

$$\lambda\rho \geq \rho^*(\mu)$$

- The bank can promise at most full disclosure in Phase B ($\rho = 1$); plugging in the post-raise belief $f(1) = 1 - b$ gives the cutoff:

$$\lambda_{\text{dyn}} = \rho^*(1 - b) = \frac{1 - b - p^*}{(1 - b)(1 - p^*)}$$

- As a result, the persuasion is sustainable if and only if

$$\lambda \geq \lambda_{\text{dyn}}$$